Steering a Paddle with the Central Complex: a Connectome-Constrained Pong Agent

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Abstract

In the fly brain, an internal estimate of heading is converted into steering commands by the central complex. A pathway of this region is isolated in the FlyWire reconstruction (156 neurons and 2190 connections, running from PFN columnar cells through hΔ interneurons to PFL3 output cells and the DNa02 descending pair) and is used unchanged as the controller of a Pong paddle. Only 141 interface parameters are optimised, by an evolution strategy, in under a minute per run. In 4 of 5 runs all 100 test matches against a fixed opponent are won, with 0.15–0.95 points conceded per match; in the remaining run most matches are lost. Because no direct input–output connection exists in the pathway, the behaviour is removed entirely when the 14 hΔ cells are silenced, which is not the case for the hemispheric output populations. A negative design result is also reported: a readout in which the output cells were pooled by hemisphere failed to learn, whereas a per-cell readout succeeded.

Keywords: central complex; PFL3; connectome; evolution strategy; closed-loop control

1 Introduction

The central complex is the insect brain region most closely associated with directed locomotion and spatial navigation. Its complete electron-microscopy connectome revealed new neuron types, sensory and motor pathways, and network motifs suited to tracking head direction and selecting actions [4]; that work followed the hemibrain reconstruction of the central fly brain [8]. Physiological studies then showed that vector arithmetic is performed in this region to compute the allocentric travelling direction [5], that three cell populations (PFL3R, PFL3L and PFL2) link the head direction system to the locomotor system [11], and that heading and goal angles are compared in a central complex circuit to produce an egocentric steering signal [6]. Descending neurons that predict and influence heading during walking, and that integrate steering signals from the head direction system with sensory ones, have also been described [7].

A controller is described by that literature, which invites a direct test: the measured circuit is handed a task whose solution is a one-dimensional steering command, and it is asked whether a minimal learned interface can make use of it. Pong is among the smallest such tasks. The paddle has one degree of freedom, the error signal is the vertical offset between ball and paddle, and a stable yardstick is provided by an opponent of adjustable strength.

Two choices keep the experiment interpretable. First, the connectome weights are never modified. Only the maps into and out of the circuit are optimised, with CMA-ES [3] rather than gradient descent, because the circuit is treated as a black box. Methods in which weights and topology are evolved together, such as NEAT [9], would alter precisely what is meant to stay fixed.

Second, abstract game readings are used instead of a simulated body; embodied fly models such as NeuroMechFly v2 [10] are a natural next step but add many free parameters. The connectivity and the transmitter predictions used here come from the public FlyWire release [1, 2].

2 Methods

Circuit. The PFN cell types PFNa (55), PFNd (40) and PFNv (21) were taken as inputs, and PFL3 (12 per hemisphere) and DNa02 (one per hemisphere) as outputs. Neurons receiving at least 20 synapses in total from the inputs and sending at least 20 to the outputs were added as intermediates (14 cells, mostly hΔA). The resulting circuit has 156 neurons and 2190 connections of at least five synapses: 1958 among the PFN population, 87 from inputs to intermediates, 70 from intermediates to outputs and 0 direct input–output connections. Every presynaptic cell is predicted to be cholinergic [2], so all weights are positive, $\log(1 + \text{synapses})$, rescaled to spectral radius 0.9 (Fig. 1).

Task. The court is 1×1. The left paddle (half-height 0.1, maximum speed 0.02 per frame) is moved by the network. The opponent tracks the ball at 0.007 per frame, but only after the ball has crossed the midline toward it, and otherwise returns to the centre. Serves start at the centre toward the network at speed 0.014 with a seeded random angle. At each paddle contact the speed is increased by 4% up to 0.028 and the outgoing angle is set from the contact offset. A match ends at 11 points or 6000 frames. The opponent speed was fixed before training: against it, almost nothing is conceded but points are rarely scored by a rule that simply follows the ball height (1.74–0.02, 62% match wins), whereas a motionless paddle loses 3.12–10.99.

Network and training. Six readings per frame (ball minus paddle height, ball position, both velocity components, paddle height, ball height) are mapped onto 16 input groups. States follow $x \leftarrow x + a(-x + g\,W \tanh(x) + I)$ with four updates per frame, and the paddle velocity is given by tanh of a trained weighted sum of output-cell activity. CMA-ES [3] was run with population 16 and four matches per candidate for 4,000,000 frames per seed (about 53 s), and repeated for 5 seeds with disjoint training, validation and test seeds.

Readout design. In a first design, the output cells were averaged within each hemisphere, mirroring the left–right organisation of the PFL3 populations [11]. Nothing was learned with it (0.59–11.00 points, the level of a motionless paddle). No improvement was obtained by varying the spectral radius or the number of updates per frame, whereas a separate readout weight for each output cell was effective (10.31–5.66). The per-cell readout (141 parameters) is used in all reported runs; the comparison is stored in `results/diagnose.csv`.

3 Results

Every test match was won in four of the five runs, with between 0.15 and 0.95 points conceded per match and at least 96.2% of incoming balls returned (Table 1). Unlike the ball-following rule, points are also scored by these agents, most plausibly through returns placed at angles the slow opponent cannot reach, although shot angles were not analysed. A weaker solution was reached in run 3 (5.55–10.89, 4% wins), which is still far above a motionless paddle. The tracking of the ball by the paddle over several rallies is shown in Fig. 2.

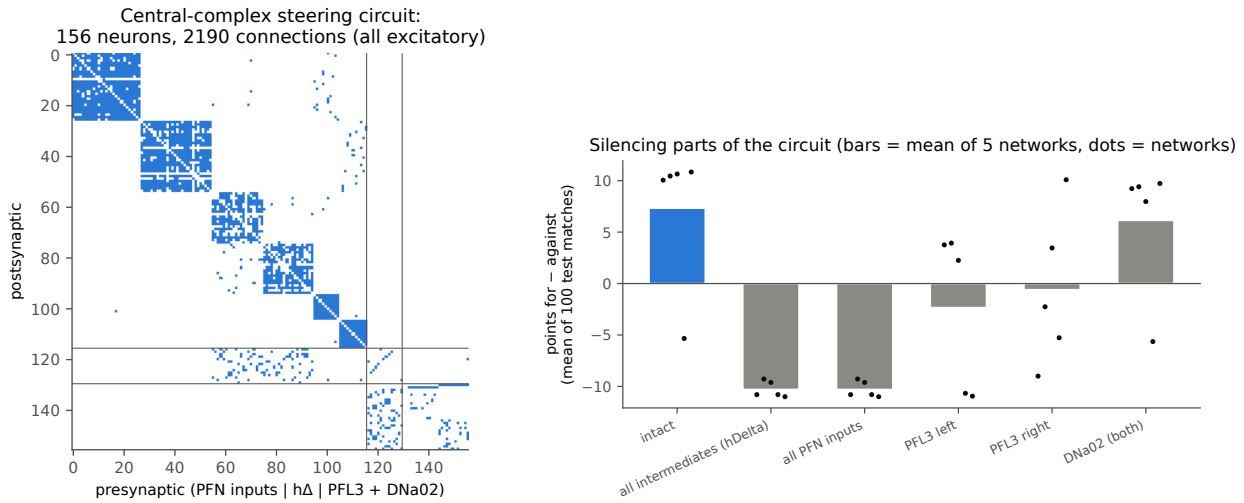


Figure 1: Left: the extracted steering pathway (PFN inputs, hΔ intermediates, PFL3 and DNa02 outputs; all connections excitatory). Right: point difference per match after silencing a group of cells; dots are the five trained networks.

Table 1: Test performance per run (100 matches each).

seed	points	match wins	returns per match	return rate	training (s)
0	11.00–0.95	100%	24.4	96.2%	52
1	11.00–0.55	100%	25.6	97.9%	56
2	10.84–0.19	100%	51.7	99.6%	52
3	5.55–10.89	4%	13.6	55.4%	53
4	10.99–0.15	100%	41.8	99.6%	54

Silencing is more consequential here than the size of the circuit suggests. When the 14 hΔ cells were removed, all matches were lost in every run (at most 1.7 points), exactly as when all PFN inputs were removed; this is expected from the absence of direct input–output connections. In contrast, some runs were left almost intact and others were broken when the PFL3 cells of one hemisphere were removed, and the removal of both DNa02 cells was mostly harmless. The learned interfaces therefore depend on the measured multisynaptic route, but the output population is used differently from run to run.

4 Discussion

A frozen central-complex pathway can serve as the controller of a one-dimensional tracking task, and the signal demonstrably passes through its interneurons. The result is bounded by three caveats. The opponent is weak by construction and the ball state is handed to the agent by the readings, so the circuit is not probed deeply by the task. Matched random graphs were not tested, so no advantage of the measured wiring is claimed; the informative comparison here is internal, between circuit parts. Finally, the readout episode is a reminder that results of this kind are partly a property of the interface: the same frozen circuit was useless or effective depending on how its output cells were pooled, which cautions against reading biological meaning into the success of such mappings. The variability across runs, with one of five failing, points the same way.

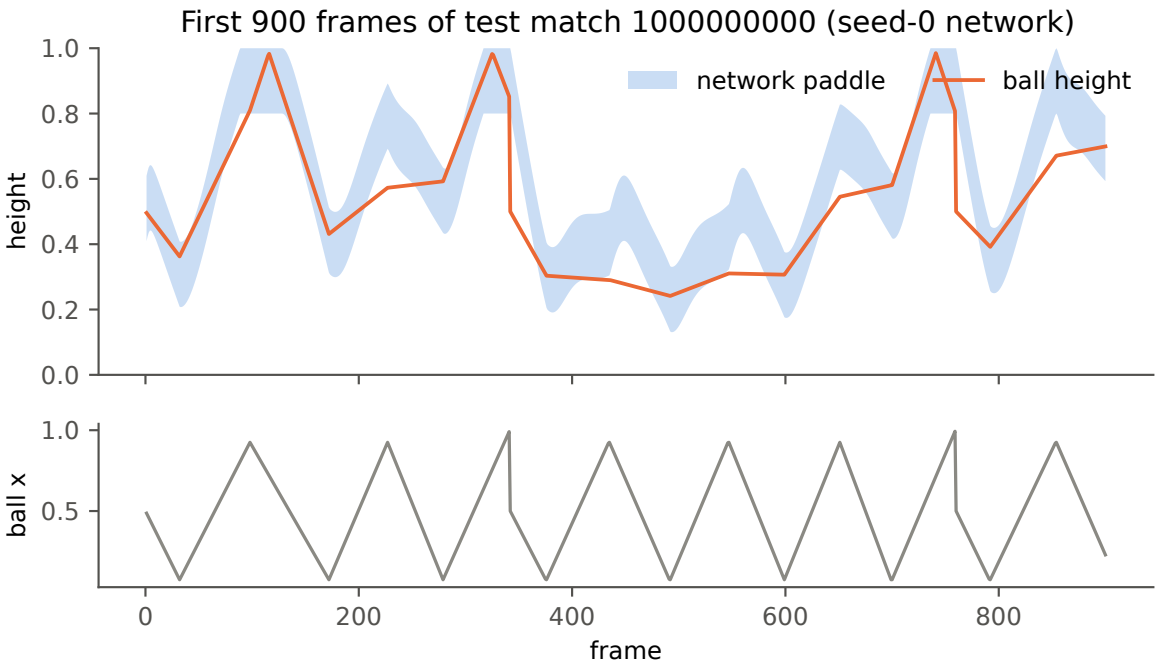


Figure 2: Ball height with the paddle's extent (top) and ball horizontal position (bottom) over the first 900 frames of a test match.

5 Conclusion

With its weights untouched, a 156-neuron steering pathway from the FlyWire connectome plays Pong well in four of five independent optimisations, and the ability is lost entirely when its hΔ interneurons are silenced.

Data and code availability. Connectivity and transmitter predictions were downloaded from the FlyWire Codex portal (v783) [1]. All scripts, per-run logs and the demo page are hosted at `https://github.com/burakgulmez/fly-pong`.

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